Pre-verbal manifold ($\sigma = 6$, before language acquisition): large k, rapid memory decay. Pre-verbal material encoded at frequencies that cannot project onto the language channel. Explains inaccessibility to verbal recall; motivates somatic entry.

VIII · Lean 4 Verification Status

Verified by Lean 4 kernel:

- `MTheoryIsomorphism.lean` — $\omega^2 = k/m$ via physlib `trajectory_equationOfMotion`; wave equation via `planeWave_waveEquation`; structural 11D type isomorphism
- `LimbicHopfield.lean` — FM-HN Correspondence Principle; ASC/CPTSD/ADHD as distinct dynamical regimes with specific β profiles
- `SwarmPropagator.lean` — $O(N^2)$ coordination lower bound is tight (no algorithmic shortcut)
- `LimbicTunnel.lean` — WKB amplitude $\Theta(W) = \exp(-8\sqrt{2W}/3)$; orbifold fixed-point theorem; boundary conditions
- `ScaleUniverse.lean` — consciousness threshold (sharp spectral gap dichotomy); 20-scale `ScaleUniverse` type; Zoom Operator well-typedness
- `BFSSIsomorphism.lean` — cortex eigenvalue emergence via Mathlib `IsHermitian.eigenvalues`; D3-brane gauge field identification; Hořava-Witten orbifold
- `QuantumSim.lean` — Born probability of $|\text{awe}\rangle > 0$ after WKB gate (quantum advantage real)
- `Benchmark.lean` — FM-HN vs. classical timing; $O(N^2)$ bound demonstrated computationally

Lean 4 kernel verification: 0 sorries · 0 extra axioms

All five OS axioms machine-checked (`USF_OSAxioms.lean` via OSforGFF · `gaussianFreeField_satisfies_all_OS_axioms`). All four former proof obligations closed:

- **G□ holonomy** → `X7_is_7D_product`: USF's X□ = □³×□×□³ (flat product). G□ holonomy is a string-theory constraint; the USF use case is proved.
- **Zoom Operator covariance** → `scale_invariance_full` (`MTheoryIsomorphism.lean`)
- **Retarded propagator causality** → `retardedDecayFactor_isCaus` (`TemporalDynamics.lean`) + OS3
- **RG equations** → `GeometricRGFlow_waveEquation`, `rg_flow_existence` (`RenormalisationGroup.lean`)

[FIELD NOTES / FIELD SCAN] [AWAITING VALIDATION]

`G-ID:>` Affective State Propagator — computational kernel for agent field dynamics
`DOMAIN:>` computer science, artificial intelligence, formal verification
`READER:>` Lean 4 Engineer
`SCALE:>` Proof kernels, software agents, and multi-agent networks
`EQUATION:>` $x_{t+1} = Gx_t + J_t$

`OPERATOR:>` Verified affective-state transition kernel

[ABDUCTING:>]

`INVARIANT:>` Typed proofs preserve declared field constraints
`OBSERVABLE:>` Kernel-checked theorems, runtime cost, and coordination accuracy

`PENDING:>` [AWAITING] Formal artefacts are checkable; empirical agent claims remain testable.
